



TECNO  
LORY

GENNAIO 2026

# REPORT CYBER- SECURITY 2026

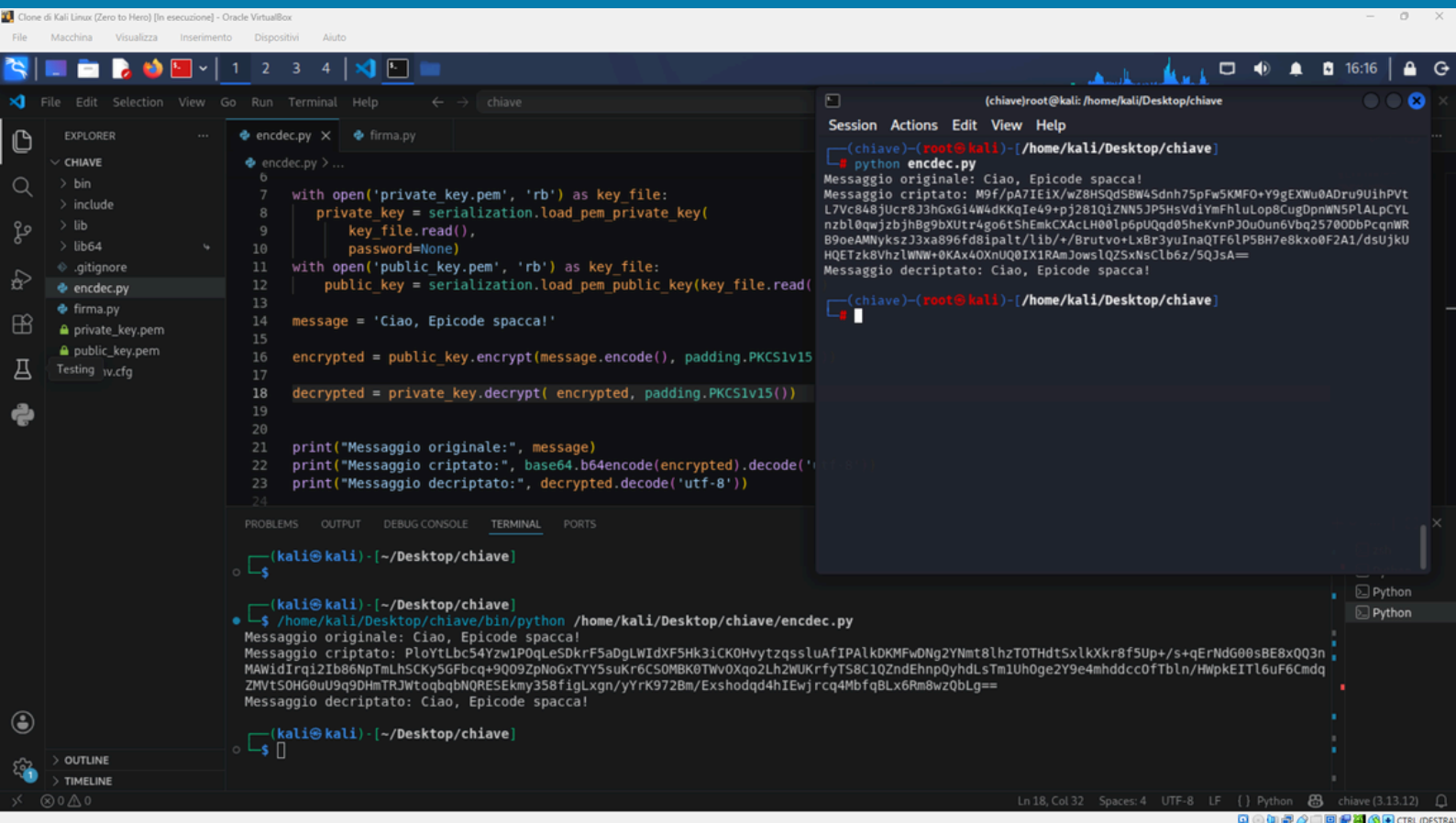
Proteggere il futuro, oggi

PRESENTATO A  
EPICODE

PRESENTATO DA  
LORENZO RIZZUTI

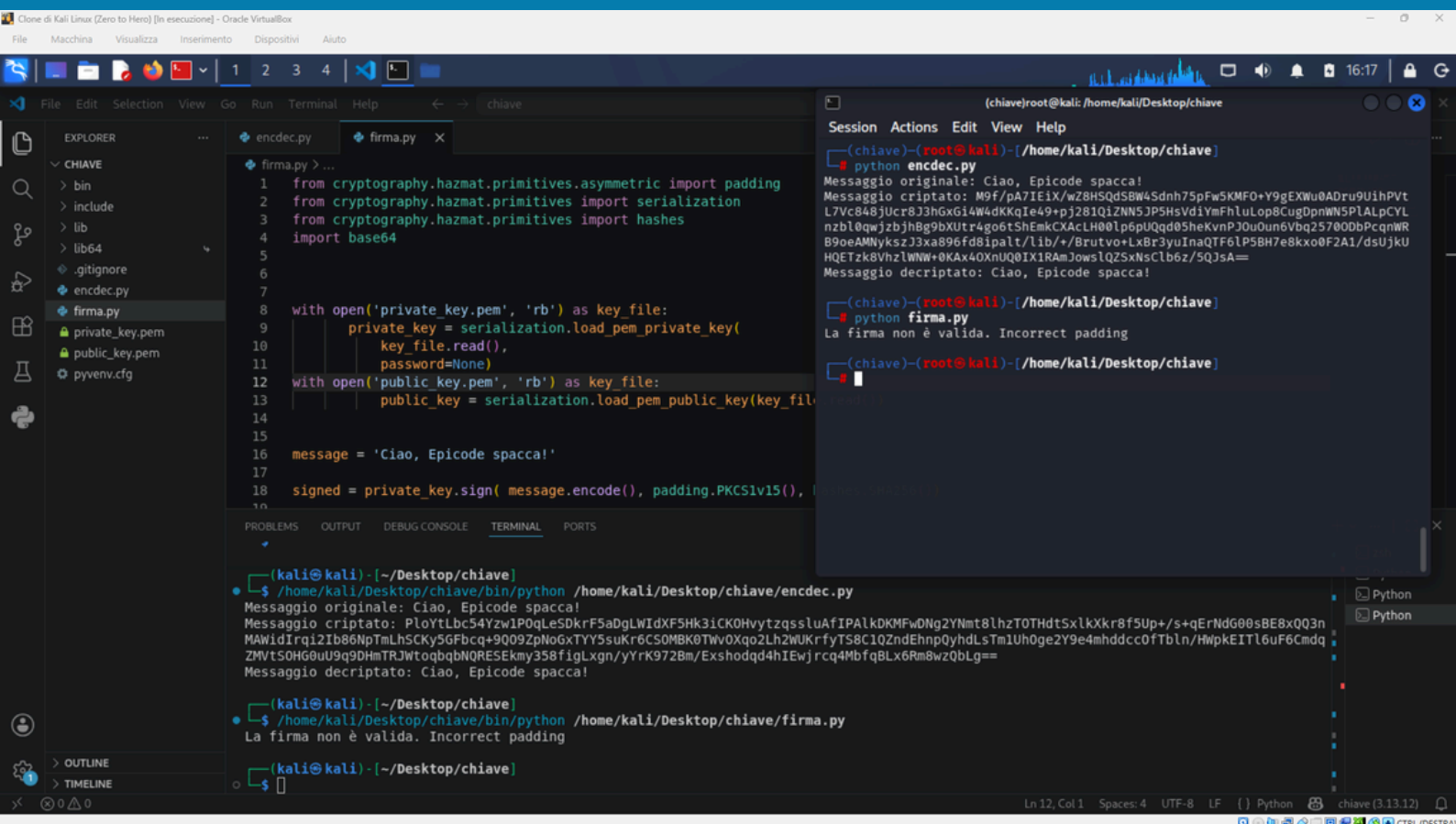
# Creazione chiave privata e chiave pubblica con OPENSSL

## Successivamente importato libreria cryptography



```
encdec.py > ...
7 with open('private_key.pem', 'rb') as key_file:
8     private_key = serialization.load_pem_private_key(
9         key_file.read(),
10         password=None)
11 with open('public_key.pem', 'rb') as key_file:
12     public_key = serialization.load_pem_public_key(key_file.read())
13
14 message = 'Ciao, Epicode spacca!'
15
16 encrypted = public_key.encrypt(message.encode(), padding.PKCS1v15())
17
18 decrypted = private_key.decrypt(encrypted, padding.PKCS1v15())
19
20
21 print("Messaggio originale:", message)
22 print("Messaggio criptato:", base64.b64encode(encrypted).decode('utf-8'))
23 print("Messaggio decrittato:", decrypted.decode('utf-8'))
24
```

```
(kali@kali) - [~/Desktop/chiave]
$ /home/kali/Desktop/chiave/bin/python /home/kali/Desktop/chiave/encdec.py
Messaggio originale: Ciao, Epicode spacca!
Messaggio criptato: PloYtlbc54Yzw1P0qLeSdKrF5aDgLWIdXF5Hk3iCK0HvytzqssluAFiPalkDKMFwDNg2YNmt8lh2T0ThtSxkXkr8f5Up+/s+qErNdG00sBE8xQ03n
MAWidIrqi2Ib86NpTmLhSCKy5Gfbcq+9Q09ZpNoGxTYYSuKr6CS0MBK0TWvOXqo2Lh2WUKrfyTS8C1QZndEhnpQyhdLsTm1Uh0ge2Y9e4mhdcc0fTbLn/HwPkEITl6uF6CmDq
ZMvTS0HG0uU9q9DHmTRJWtoqbqBNQRESEkmy358figLxgn/yYrk972Bm/Exshodqd4hIEwjrCq4MbfgBLx6Rm8wQbLg==
Messaggio decrittato: Ciao, Epicode spacca!
```



```
firma.py > ...
1 from cryptography.hazmat.primitives.asymmetric import padding
2 from cryptography.hazmat.primitives import serialization
3 from cryptography.hazmat.primitives import hashes
4 import base64
5
6
7
8 with open('private_key.pem', 'rb') as key_file:
9     private_key = serialization.load_pem_private_key(
10         key_file.read(),
11         password=None)
12 with open('public_key.pem', 'rb') as key_file:
13     public_key = serialization.load_pem_public_key(key_file.read())
14
15
16 message = 'Ciao, Epicode spacca!'
17
18 signed = private_key.sign(message.encode(), padding.PKCS1v15(), hashes.SHA256())
19
```

```
(kali@kali) - [~/Desktop/chiave]
$ /home/kali/Desktop/chiave/bin/python /home/kali/Desktop/chiave/encdec.py
Messaggio originale: Ciao, Epicode spacca!
Messaggio criptato: PloYtlbc54Yzw1P0qLeSdKrF5aDgLWIdXF5Hk3iCK0HvytzqssluAFiPalkDKMFwDNg2YNmt8lh2T0ThtSxkXkr8f5Up+/s+qErNdG00sBE8xQ03n
MAWidIrqi2Ib86NpTmLhSCKy5Gfbcq+9Q09ZpNoGxTYYSuKr6CS0MBK0TWvOXqo2Lh2WUKrfyTS8C1QZndEhnpQyhdLsTm1Uh0ge2Y9e4mhdcc0fTbLn/HwPkEITl6uF6CmDq
ZMvTS0HG0uU9q9DHmTRJWtoqbqBNQRESEkmy358figLxgn/yYrk972Bm/Exshodqd4hIEwjrCq4MbfgBLx6Rm8wQbLg==
Messaggio decrittato: Ciao, Epicode spacca!

(kali@kali) - [~/Desktop/chiave]
$ /home/kali/Desktop/chiave/bin/python /home/kali/Desktop/chiave/firma.py
La firma non è valida. Incorrect padding
```

